

Technical Task

Software Engineer (Full-stack)

Insightful text analysis: Product overview

Mitabits Insightful text analysis helps to derive the insights and meaning of the text. It enables users to analyse text and extract information about people, organisation, job roles, and better understand temporal information.

More specifically, it is a web application where user can put any text and instantly get information about the extracted mentions of various entities in the knowledge base.

Text analysis

Alan Turing was born on 23/06/1912. During the Second World War, he worked as a scientist for the Government Code and Cypher School (currently GCHQ) at Bletchley Park, Britain's codebreaking centre that produced Ultra intelligence.

Analyze

Type	Content	Meta
DATE	"23/06/1912"	1912-06-23
PERSON	"Alan Turing"	Lookup
ORGANISATION	"GCHQ"	Lookup

Figure 1: Text analysis

In addition, user can search for a specific person name or organisation in the knowledge base to get more detailed information about them.

Knowledge Base Search

Person

Search

[Try Alan Turing](#) or [Google](#)

Alan Turing 100% match

synonyms: Alan Mathison Turing

dob: 1912-06-23

bio: Alan Mathison Turing is an English mathematician, computer scientist, logician, cryptanalyst, philosopher, and theoretical biologist. Turing was highly influential in the development of theoretical computer science, providing a formalisation of the concepts of algorithm and computation with the Turing machine, which can be considered a model of a general-purpose computer. Turing is widely considered to be the father of theoretical computer science and artificial intelligence.

Figure 2: Knowledge Base Search

Task

Your task is to explore and then work with the existing product codebase consisting of the React front-end application and the natural language API (Python, Flask) to complete the following subtasks.

We appreciate your free time is important. Therefore, spend around two hours in total or until you feel that your solution demonstrates enough for us to assess the application. If you have unfinished parts, please leave some notes in your report explaining what would you do next given more time. We would be happy to review this.

1 Existing codebase review

Review the codebase and make notes about potential issues - bugs, scalability and performance issues and things to consider in the future (one or two paragraphs is enough).

[15 mins]

2 Brief system architecture diagram

Very briefly draw the system architecture diagram outlining internals of different components and interactions between them.

It is up to you to decide which level of details to maintain, but generally it should be sufficient to provide a brief overview of the system to another developer. You can use your favourite diagramming tool or simply a pen and paper.

[15 mins]

3 Implement entity recognition for roles

Currently, mentions of job roles (such as scientist, software developer) cannot be recognised. Your task is to add this functionality to the natural language API by implementing `RoleAnnotator` component.

See `resources/roles.yml` for the list of available roles that your algorithm must recognise.

[30 mins]

4 Highlight entities in text

You are asked to have a look into the following user story:

As a user who works with long documents I want to see recognised mentions directly in the document as highlighted text spans so that there will be no need to scroll up and down checking the table with recognised mentions.

Implement this functionality considering user experience and make notes about any assumptions you've made.

[30 mins]

Development Environment Setup

You have been given access to two repositories - `natural-language-api` (backend) and `text-analysis-app` (frontend). See `README.md` for more details and instructions.

Assessment

We will assess the task based on the following criteria:

- Problem solving and decision making
- How clean, modular and extensible the code is
- Testing and considering edge cases in your solutions
- Anything above and beyond

Submission

Please send your report, source code and supporting documentation to info@mitabits.com.

Please do not hesitate and contact us at info@mitabits.com if you have any questions.

Good luck!

Mitabits Team